

Unit 9

Similar Figures

Sr. No.	Questions	A	B	C	D
1	If two polygons are similar, then:	their corresponding angles are equal. ✓	their areas are equal.	their volumes are equal.	their corresponding sides are equal.
2	The ratio of the areas of two similar polygons is:	equal to the ratio of their perimeters.	equal to the square of the ratio of their corresponding sides. ✓	equal to the cube of the ratio of their corresponding sides.	equal to the sum of their corresponding sides.
3	If the volume of two similar solids is 125 cm^3 and 27 cm^3 , the ratio of their corresponding heights is:	3:5	5:3 ✓	25:9	9:25
4	The exterior angle of a regular pentagon is:	40°	45°	60°	72° ✓
5	A parallelogram has an area of 64 cm^2 and a similar parallelogram has an area of 144 cm^2 . If a side of the smaller parallelogram is 8 cm , the corresponding side of the larger parallelogram is:	10 cm	12 cm ✓	18 cm	16 cm
6	The total number of diagonals in a polygon with 9 sides is:	18	21	25	27 ✓
7	Two spheres are similar, and their radii are in the ratio 4: 5. If the surface area of the larger sphere is $500\pi \text{ cm}^2$, what is the surface area of the smaller sphere?	$256\pi \text{ cm}^2$	$320\pi \text{ cm}^2$ ✓	$400\pi \text{ cm}^2$	$405\pi \text{ cm}^2$
8	A regular polygon has an exterior angle of 30° . How many diagonals does the polygon have?	54 ✓	90	72	108
9	In a regular hexagon, the ratio of the length of a diagonal to the side length is:	$\sqrt{3}:1$	$2:1$ ✓	3: 2	2: 3
10	A regular polygon has an interior angle of 165° . How many sides does it have?	15	16	20	24 ✓

Solution of MCQs

- 1 Similar polygons have corresponding angles equal and corresponding sides proportional.

2	The area ratio of similar polygons is the square of the ratio of their corresponding sides.
3	$\frac{V_1}{V_2} = \left(\frac{l_1}{l_2}\right)^3$ $\Rightarrow \frac{125}{27} = \left(\frac{l_1}{l_2}\right)^3$ $\sqrt[3]{\frac{125}{27}} = \sqrt[3]{\left(\frac{l_1}{l_2}\right)^3}$ $\frac{5}{3} = \frac{l_1}{l_2}$ $\frac{l_1}{l_2} = \frac{5}{3}$
4	Exterior angle of pentagon is $= \frac{360^\circ}{5} = 72^\circ$
5	Ratio of areas of similar figures $\frac{A_1}{A_2} = \left(\frac{l_1}{l_2}\right)^2$ $\frac{64}{144} = \left(\frac{8}{l_2}\right)^2$ $\sqrt{\frac{64}{144}} = \sqrt{\left(\frac{8}{l_2}\right)^2}$ $\frac{8}{12} = \frac{8}{l_2}$ $l_2 \times 8 = 12 \times 8$ $l_2 = \frac{96}{8}$ $l_2 = 12 \text{ cm}$
6	The total number of diagonals in a polygon with 9 sides is $= \frac{n(n-3)}{2} = \frac{9(9-3)}{2} = \frac{54}{2} = 27$
7	Ratio of areas of similar figures is $\frac{A_1}{A_2} = \left(\frac{l_1}{l_2}\right)^2$ $\frac{A_1}{500\pi} = \left(\frac{4}{5}\right)^2$ $\frac{A_1}{500\pi} = \frac{16}{25}$ $A_1 \times 25 = 16 \times 500\pi$ $A_1 = \frac{16 \times 500\pi}{25}$ $A_1 = 320\pi$
8	Exterior angle of a regular polygon is given $\text{Exterior Angle} = \frac{360^\circ}{n}$ $30^\circ = \frac{360^\circ}{n}$ $n = \frac{360^\circ}{30^\circ}$ $\text{no. of sides} = 12$ Now the total number of diagonals in a polygon with 12 sides is $= \frac{n(n-3)}{2} = \frac{12(12-3)}{2} = \frac{108}{2} = 54$
9	In a regular hexagon, the longest diagonal connects opposite vertices and is twice the side length , so the ratio is 2: 1.

10	Exterior angle = $180^\circ - 165^\circ = 15^\circ$ $\text{Exterior Angle} = \frac{360^\circ}{n}$ $15^\circ = \frac{360^\circ}{n}$ $n = \frac{360^\circ}{15^\circ}$ <i>no. of sides</i> = 24
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